

AutoSeg Evaluator User Manual

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1. Introduction and Disclaimer

AutoSeg Evaluator is an open-source application developed to support efficient evaluation of AI-generated contours and comparison of contours from multiple automated or manual sources.

The software facilitates evidence generation to support the evaluation, acceptance testing, commissioning, and ongoing quality assurance of auto-segmentation systems. Its outputs must be interpreted by appropriately qualified radiation oncology professionals alongside expert contour review, relevant clinical guidance, and local commissioning requirements. Software outputs must not be used as the sole basis for clinical decisions or approval of an auto-segmentation system for clinical use. Responsibility for interpreting the findings and determining suitability for local clinical practice remains with the commissioning team, relevant multidisciplinary professionals, and the institution.

AutoSeg Evaluator does not establish clinical acceptability or prescribe universal acceptance thresholds. Metric values and statistical comparisons alone are insufficient to determine whether a contour is suitable for its intended clinical use. The software supports evidence generation and professional assessment; it does not replace clinical judgement.

2. Installing and Getting Started

AutoSeg Evaluator runs from a portable download. Nothing is installed, and it needs no Python, no administrator rights and no internet connection.

	Windows	Linux
System	Windows 10 or 11, 64-bit	64-bit (x86-64), glibc 2.34 or newer: Ubuntu 22.04, Debian 12, RHEL 9 or later
Download	about 390 MB	about 440 MB

The Windows download takes about 1.2 GB once extracted. On macOS, run it from source (section 2.4).

2.1 Windows

1. Download **AutoSegEvaluator-v3.0.1.zip** from the [Releases page](#).
2. Extract it to a folder with a short path, such as C:\AutoSegEvaluator.
 - Windows limits a file's full path to 260 characters, and the app's files sit deep inside its folder. The folder's own path must therefore be about 80 characters or fewer. From deeper than that, AutoSeg Evaluator cannot load all its files.
3. Double-click Run AutoSeg Evaluator.bat. For a desktop icon, also run Create Desktop Shortcut.vbs.

2.2 Linux

1. Download **AutoSegEvaluator-v3.0.1-linux-x86_64.tar.gz** from the [Releases page](#).
2. Extract it: `tar -xzf AutoSegEvaluator-v3.0.1-linux-x86_64.tar.gz`
3. Run `./run-autoseg-evaluator.sh`. To add the app to your applications menu, run `./add-to-applications-menu.sh`.

If it reports that the Qt platform plugin "xcb" could not be loaded, install the missing libraries:

- Ubuntu / Debian: `sudo apt install libxcb-cursor0 libxkbcommon-x11-0`
- Fedora / RHEL: `sudo dnf install xcb-util-cursor libxkbcommon-x11`

2.3 MacOS

You need Python 3.10 or newer. Follow the source-install steps in the project README.

2.4 The 2D Metrics Engine

The 2D contour metrics (added path length, 2D Hausdorff distance and contour distances) are computed by one of two engines.

- **Fast engine:** a compiled library, included in the Windows and Linux downloads and used automatically.
 - The Windows library is the supplier's validated build.
 - The Linux library was built from the same source code and passed the supplier's acceptance tests on three Linux versions, with the same results as the Windows build.
- **Reference engine:** plain Python, used automatically wherever no compiled library exists, such as on macOS or ARM computers. It uses the same definitions and agrees with the fast engine to within 0.02 mm. It is roughly 60–150 times slower, and the largest structures, such as a body contour, may not complete.

The Compute Tab shows which engine will run, for example, the fast engine will display as *Engine: fast 0.2.0.dev2*.

Advanced: to force the reference engine, for example as an independent cross-check, set the environment variable `AUTOSEG_POLYGON_ENGINE=reference` before starting the app.

2.5 Your Data and Files

- Your data stays on your computer. AutoSeg Evaluator makes no network connections.
- `settings.json` in the app's folder stores your preferences, so keep the folder somewhere you can write to.
- `synonyms.json` is optional. Placed beside the launcher, it replaces the built-in TG-263 synonym dictionary.
- Sessions, exports and reports are saved wherever you choose.

2.6 Updating

Extract the new version to a new folder. Copy `settings.json`, and `synonyms.json` if you made one, from the old folder to keep your preferences.

3. End to End workflow

This section demonstrates how to create a cohort from DICOM files and generate a report with 3D, 2D and dose metrics. It uses five tabs: 1 Load Data → 3 Match Contours → 5 Compute → 6 Results → 7 Report. Tabs 2 and 4 are optional and covered in Chapters 4 and 5.

3.1 Before you Start

Put everything in one folder; sub-folders are searched too. For each patient you need:

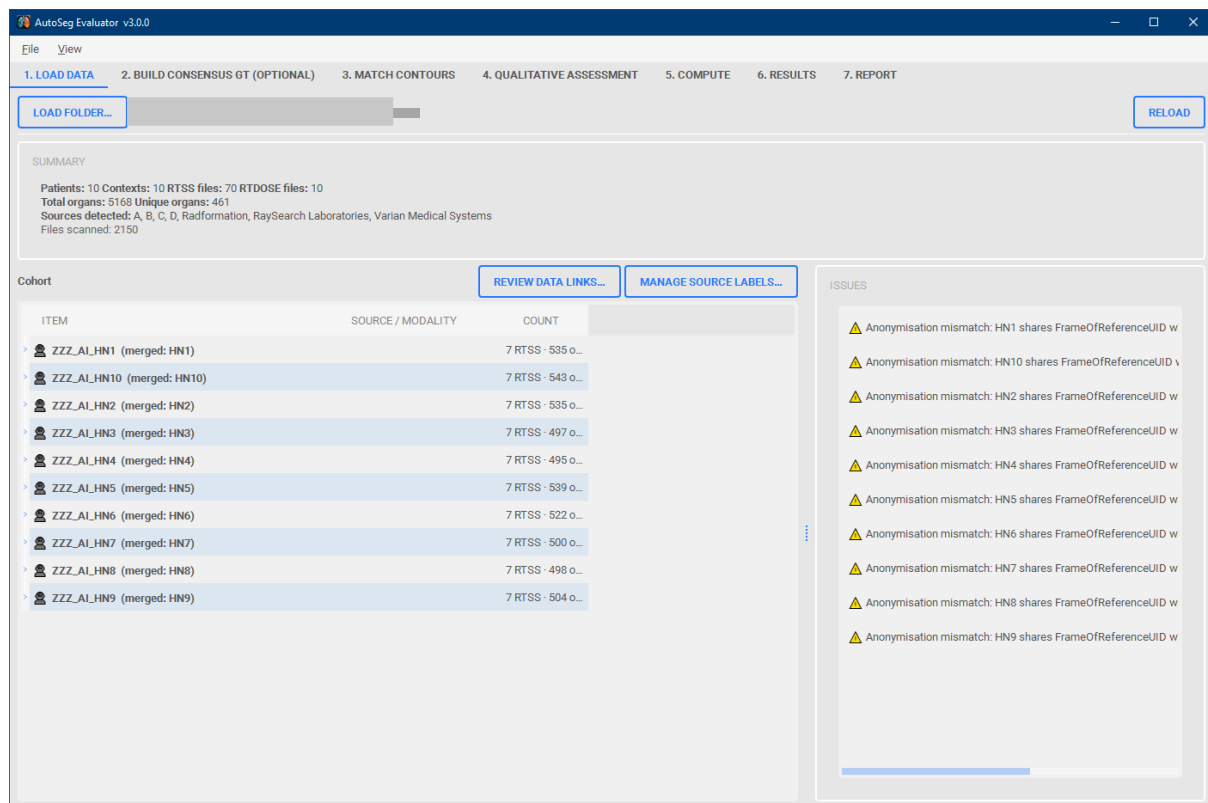
- the CT images,
- an RT Structure Set from each source: the ground truth plus every system or observer you want to test,

- an RT Dose, if you want dose metrics.

No RT Plan is needed.

3.2 Load Data (Tab 1)

- 1) Open AutoSeg Evaluator and select **Load Folder...**
- 2) Choose your main folder containing one or more patient's worth of DICOM CT, RTSS, DICOM Dose files.



- 3) Check the source labels for each vendor or manual observer. Each structure set is labelled with the system or observer it came from, read from its DICOM tags. If a label is unhelpful or you wish to change it:
 - a. Click **Manage Source Labels...**
 - b. Double-click a cell in the **New Source Label** column and type a label such as "Vendor A" or "Observer A".
 - c. To label many files at once, bulk-select rows by:
 - i. Highlight a single row and press **Select Similar...** and find your preferred grouping. Then type the label in the Bulk Apply section and click **Apply To Selected**.
 - ii. Or, press **Group by Column**, find your preferred grouping using the dropdown menu, and type your preferred source label for each unique source by double-clicking the **New Source Label** column.
 - d. Press Apply to save the custom labels. These are remembered across sessions.

Manage Source Labels

Override the detected source label for any RTSTRUCT below. Custom labels persist across sessions and apply everywhere the source is shown (drawers, results CSV, etc.). Leave the field blank to keep the auto-detected value. *Tip: for multi-observer studies, give each observer a distinct label (e.g. 'Observer A'). Group by column partitions every file into observer groups in one step; Select similar is a per-file fallback. Right-click any column header to show or hide columns.*

FILE	ATTEN	DETECTED SOURCE	ORIGIN	MANUFACTURER	STRUCTURESETLABEL	SOFTWAREVERSIONS	STRUCTURESETNAME	STRUCTURESETDESC
STRUCT_A.dcm	ZZZ_AI_...	A	Manufacturer tag	A	A	7.4.2	A	
STRUCT_B.dcm	ZZZ_AI_...	B	Manufacturer tag	B	B	system:1.2.7,worker.bccf05...		B
STRUCT_C.dcm	ZZZ_AI_...	C	Manufacturer tag	C	C	1.8.0	C	C
STRUCT_D.dcm	ZZZ_AI_...	D	Manufacturer tag	D	D	v2.3.1	D	
HN1 Man_fix2.dcm	ZZZ_AI_...	Varian Medical Systems	Manufacturer tag	Varian Medical Systems		16.1.4		
	ZZZ_AI_...	Radformation	Manufacturer tag	Radformation	AC_Head_and_Neck	2.7.2.0		
	ZZZ_AI_...	RaySearch Laboratories	Manufacturer tag	RaySearch Laboratories	RS: Unapproved	16.0.0.847 (Dicom Export)		
STRUCT_A.dcm	ZZZ_AI_...	A	Manufacturer tag	A	A	7.4.2	A	
STRUCT_B.dcm	ZZZ_AI_...	B	Manufacturer tag	B	B	system:1.2.7,worker.bccf05...		B
STRUCT_C.dcm	ZZZ_AI_...	C	Manufacturer tag	C	C	1.8.0	C	C
STRUCT_D.dcm	ZZZ_AI_...	D	Manufacturer tag	D	D	v2.3.1	D	
HN10 Man_fix2.dcm	ZZZ_AI_...	Varian Medical Systems	Manufacturer tag	Varian Medical Systems		16.1.4		

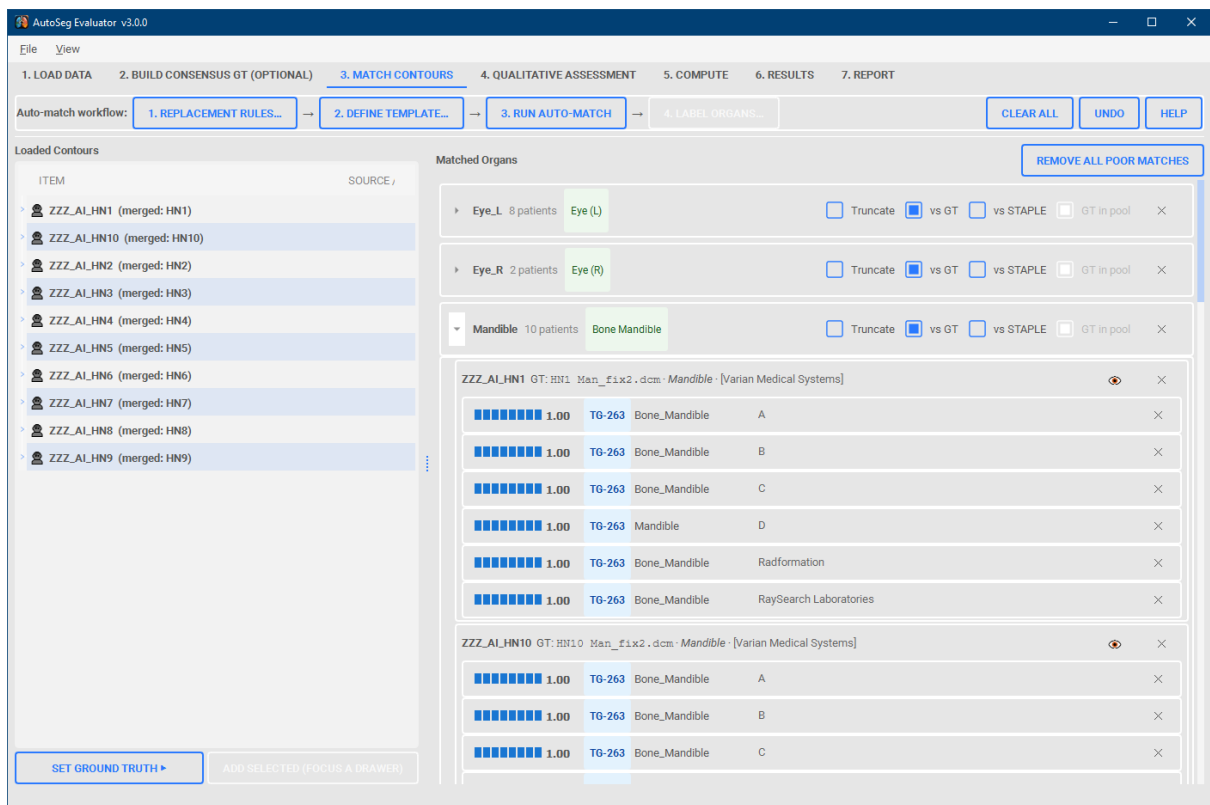
Bulk apply:

- Lastly, settle any data link issues. If the **Review Data Links...** button shows a number, open it. Each structure set is normally linked to its CT and dose automatically; this dialog lists any it couldn't decide, so you can choose. Compute won't run until they're settled. Unsettled dose links only block a run that includes dose metrics.

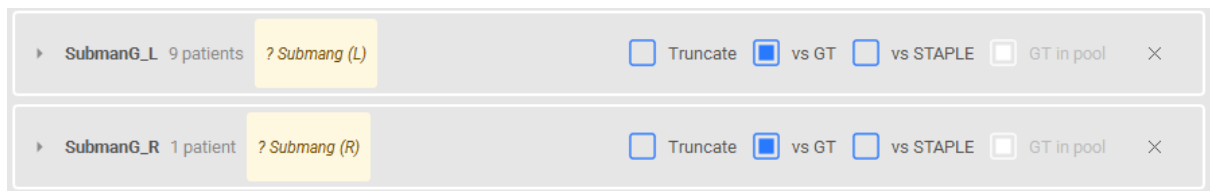
3.3 Match Contours (Tab 3)

This tab decides, organ by organ, which contour is the ground truth and which contours are tested against it. Work through the numbered buttons along the top:

- Replacement Rules...** (optional): fix local naming quirks, for example Musc_Constrict → Pharyngeal_Constrictor
- Define Template...**:
 - List the ground truth contours, one per line
 - Under **Source Label Contains**, enter your ground truth's source label
 - Set the string similarity threshold for non-TG263 detected structures: test matches scoring below it are flagged . The default is 0.60.
- Run **Auto Match**
 - This creates one **drawer** per detected organ. Each drawer holds the ground truth for every patient, plus the matched test contours with a similarity score.
- Review the drawers
 - Remove All Poor Matches** button clears every flagged row; removes a single entry.
 - Drag a missing contour from Loaded Contours into a drawer.
 - Click the eye icon to view a patient's contours, with the dose overlaid if one is loaded.
 - Undo (Ctrl+Z) reverses up to 20 changes.



- 5) For any ground truth organ not recognized as a TG263 organ, the drawer displays the name in a yellow box. These drawers must be assigned a canonical organ name in order to properly compute statistics in the Report tab.



- 6) Click the Label Organs... button and either accept or change the automatically suggested TG263 organ. Upon pressing **Ok**, the organ now turns green suggesting it is a recognized organ.

Label Organs

These drawers have ground-truth names the TG-263 dictionary does not recognise, so each is currently its own organ and pools with nothing. Saying what they are lets results from differently-named drawers be analysed together.

Nothing is merged, renamed or moved. This attaches a label; the drawers stay exactly as they are.

Only organs are listed — targets, couch, rings, dose levels and PRVs are already classified and are left alone.

Filter drawers...

NO SUGGESTIONS PENDING

DRAWER	PATIENTS	GROUND-TRUTH NAME(s)	IS THIS ORGAN...	SUGGESTED FROM
SubmanG_L	9 patients	SubmanG_L	✓ GInd Submand (45/45 of its contours)	GInd_Submand_L
SubmanG_R	1 patient	SubmanG_R	✓ GInd Submand (5/5 of its contours)	GInd_Submand_R

2 unrecognised drawer(s) · 2 labelled. Leaving one unlabelled is fine — it simply stays its own organ.

OK

CANCEL

SubmanG_L 9 patients

GInd Submand (L)

☐ Truncate
☒ vs GT
☐ vs STAPLE
☐ GT in pool

×

SubmanG_R 1 patient

GInd Submand (R)

☐ Truncate
☒ vs GT
☐ vs STAPLE
☐ GT in pool

×

Drawer Options

- **vs GT** (on by default): compare each test contour with the ground truth.
- **Truncate**: ignore test slices beyond the ground truth's top and bottom. This helps for structures a system may legitimately draw further, such as the spinal cord or rectum.
- **vs STAPLE** and **GT in pool** are covered in Section 4.

3.4 Compute (Tab 5)

Select the desired metrics to compute.

Group	Metrics	Also enter
3D mask metrics	Dice · Precision + recall · 3D Hausdorff (100%) and (95%) · Mean Surface Distance · Surface Dice · Volume (cc) + diff/ratio · Centre-of-mass offset	Surface Dice τ (mm)
2D contour metrics	Added Path Length · Normalised APL · 2D Hausdorff (100%) and (95%) · Mean and Median contour distance	APL tolerance τ (mm)
Dose (DVH)	Dmean · Dmax · Dmin	D at volume (%), D at volume (cc), V at dose (Gy), as comma lists, e.g. 95, 50, 2

- **Tolerances**: one value, or several (1, 2, 3) to get a column for each.
- **Dose**: found automatically for each patient, preferring a plan dose.
- **Contour checks** always run, whatever you tick:
 - overlap with the PTV, using contours with structure type PTV in the ground-truth structure set,
 - Contour Discontinuity, for test contours that skip a slice.

- **Record audit detail** (optional) keeps the full detail behind every number. Tick it before the run; it can't be added afterwards.
- **Metric definitions...** explains exactly how each metric is computed.

Click **Compute All**. If the run can't start, the tab says why: no metric selected, a missing tolerance, or unsettled data links.

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File View

1. LOAD DATA 2. BUILD CONSENSUS GT (OPTIONAL) 3. MATCH CONTOURS 4. QUALITATIVE ASSESSMENT **5. COMPUTE** 6. RESULTS 7. REPORT

3D MASK METRICS (RASTERISED)

Contours are filled onto the CT voxel grid and the resulting binary masks are compared in 3D.

☐ Dice

☐ Precision + recall

☐ 3D Hausdorff (100%)

☐ 3D Hausdorff (95%)

☐ Mean Surface Distance

☐ Surface Dice

☐ Volume (cc) + diff/ratio

☐ Centre-of-mass offset

Surface Dice τ (mm): 3 or 1, 2, 3

2D CONTOUR METRICS (NATIVE RTSS POLYGONS)

Measured on the contour line segments as stored – no voxels, no sampling. Distances use only the planes both structures reach.

☐ Added Path Length

☐ Normalised APL

☐ 2D Hausdorff (100%)

☐ 2D Hausdorff (95%)

☐ Mean contour distance

☐ Median contour distance

APL tolerance τ (mm): 3 or 1, 2, 3

Engine: fast 0.2.0.dev2

DOSIMETRIC METRICS (DVH)

RT Dose: auto-detected per patient from the loaded folder (prefers DoseSummationType = PLAN).

☐ Dmean ☐ Dmax ☐ Dmin

D at volume (%): 95, 50, 5, 2

D at volume (cc): 0.1, 1, 2

V at dose (Gy): 20, 30, 40

D_x%: dose to the hottest X% of the structure (Gy). D_{1cc}: dose to the hottest 1% of the structure (Gy). V_xGy: volume (cc) receiving $\geq$ X Gy. Examples: D95, D50, D2 – D0.1cc, D1cc, D2cc – V20Gy, V30Gy.

10 RTDOSE file(s) detected across 10/10 patient(s).

STAPLE CONSENSUS PARAMETERS

3.5 Results (Tab 6)

- One row per test contour (patient, organ, source).
- Columns are grouped in coloured bands: identifiers, overlap, surface distances, 2D metrics, volume, contour checks and dose. Hover over a band for its description. Click a column header to sort.
- Where a metric can't be measured, the table says why instead of showing a zero.
- **Export CSV...** saves the table. If audit detail was recorded, a matching .audit.json file is saved beside it.

AutoSeg Evaluator v3.0.0											
File View											
1. LOAD DATA 2. BUILD CONSENSUS GT (OPTIONAL) 3. MATCH CONTOURS 4. QUALITATIVE ASSESSMENT 5. COMPUTE 6. RESULTS 7. REPORT											
Results										280 rows	
CLEAR EXPORT CSV...											
DRAWER	ORGAN	SIDE	STRUCTURE TYP	PATIENT	MODE	DESIGNATED GT	GT SOURCE	GT RTSS	GT ROI	TEST SOURCE	
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt_dose	True	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	Varian Medical Systems	HN1 I
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	A	STRU
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	B	STRU
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	C	STRU
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	D	STRU
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	Radformation	AC-2
Eye_L	Eye (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Eye_L	RaySearch Laboratories	RS1.2
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt_dose	True	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	Varian Medical Systems	HN1 I
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	A	STRU
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	B	STRU
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	C	STRU
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	D	STRU
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	Radformation	AC-2
Mandible	Bone Mandible		oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Mandible	RaySearch Laboratories	RS1.2
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt_dose	True	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	Varian Medical Systems	HN1 I
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	A	STRU
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	B	STRU
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	C	STRU
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	D	STRU
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	Radformation	AC-2
Parotid_R	Parotid (R)	R	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	Parotid_R	RaySearch Laboratories	RS1.2
SubmanG_L	GlnD Submand (L)	L	oar	ZZZ_AI_HN1	gt_dose	True	Varian Medical Systems	HN1 Man_fix2.dcm	SubmanG_L	Varian Medical Systems	HN1 I
SubmanG_L	GlnD Submand (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	SubmanG_L	A	STRU
SubmanG_L	GlnD Submand (L)	L	oar	ZZZ_AI_HN1	gt	False	Varian Medical Systems	HN1 Man_fix2.dcm	SubmanG_L	B	STRU

3.6 Report (Tab 7)

Set up the comparison:

- **Ground truth:** what every metric was measured against.
- Compare across:
 - *Organs* — one challenger compares one source with the reference across several organs;
 - *Sources* — one organ compares every source with the reference on one organ.
- **Metric, Compare against** (the reference source) and **Challenger**.
- **Organs:** select the organs to show. Each is analysed on its own.

The page then shows:

- Coverage: who contoured what, plus the truncation, PTV-overlap and discontinuity counts.
- Descriptive statistics for each organ and source.
- Paired comparison: Wilcoxon signed-rank test and Hodges–Lehmann difference with its 95% interval, using only patients contoured by both sources.
- Paired differences, one line per patient.
- Difference from reference, a forest plot.
- Acquisition parameters of the CT images.

Export PDF... saves the whole page as a report. **Save figures...** saves each figure as a 200-dpi PNG.

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File View

1. LOAD DATA 2. BUILD CONSENSUS GT (OPTIONAL) 3. MATCH CONTOURS 4. QUALITATIVE ASSESSMENT 5. COMPUTE 6. RESULTS 7. REPORT

Statistical Report 10 patients · 7 organs · 6 sources · vs Varian Medical Systems

SAVE FIGURES... EXPORT PDF...

COMPARISON

Ground truth: Varian Medical Systems

Compare across: Organs — one challenger

Metric: dice

Compare against: A

Challenger: C

Organs in the correction family: Bone Mandible, Eye (L), Eye (R)

ALL NONE

3 of 7 organs have fewer than six paired patients, which cannot produce a p-value at or below 0.05 whatever the data show. Their p-values are not evidence of similarity.

7 comparisons are shown, each answering its own question. p-values are per organ and unadjusted, so a row does not change because another is displayed. If you scan the table for rows below 0.05, expect about 0.4 to appear there by chance even if every source performed identically — so report every row, not only the ones that crossed.

The rows do not all use the same patients. Each comparison runs on the patients that pair for it, so a row is sound on its own but two rows are not measured on the same cohort. The paired view below always shows the patients belonging to the row it names.

COVERAGE

ORGAN	SOURCE	COVERAGE	TRUNCATED	TEST-ONLY PTV OVERLAP	CONTOUR DISCONTINUITY
Bone Mandible	A	10 / 10	no	0 of 10	0 of 10
	B	10 / 10		0 of 10	0 of 10
	C	10 / 10		0 of 10	0 of 10
	D	10 / 10		0 of 10	0 of 10
	Radformation	10 / 10		0 of 10	0 of 10
	RaySearch ...	10 / 10		0 of 10	0 of 10
Eye (L)	A	8 / 8	no	1 of 8	0 of 8
	B	8 / 8		1 of 8	0 of 8
	C	8 / 8		0 of 8	0 of 8

3.7 Save Session

File → Save Session stores everything: the folder, rules, template, drawers, labels, data-link choices and computed results.

File → Load Session... picks up where you left off.

A session stores the data folder's location, not the files themselves, so keep the data where it was.

4. STAPLE Workflows

STAPLE combines several segmentations of the same structure into one consensus, giving each rater a weight based on how well it agrees with the others. AutoSeg Evaluator offers it in two ways:

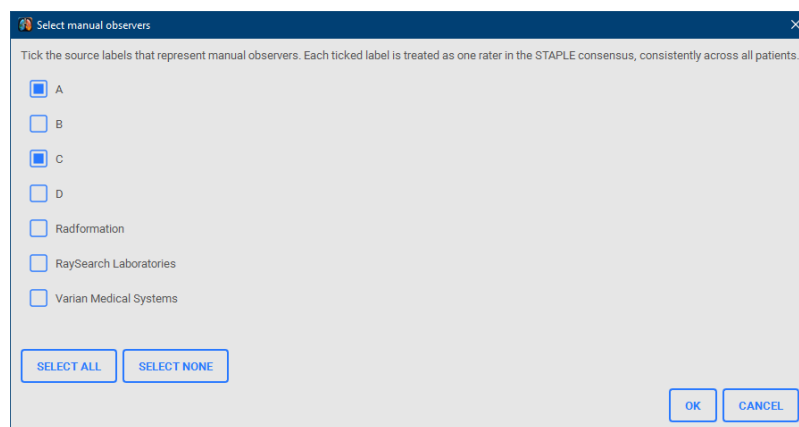
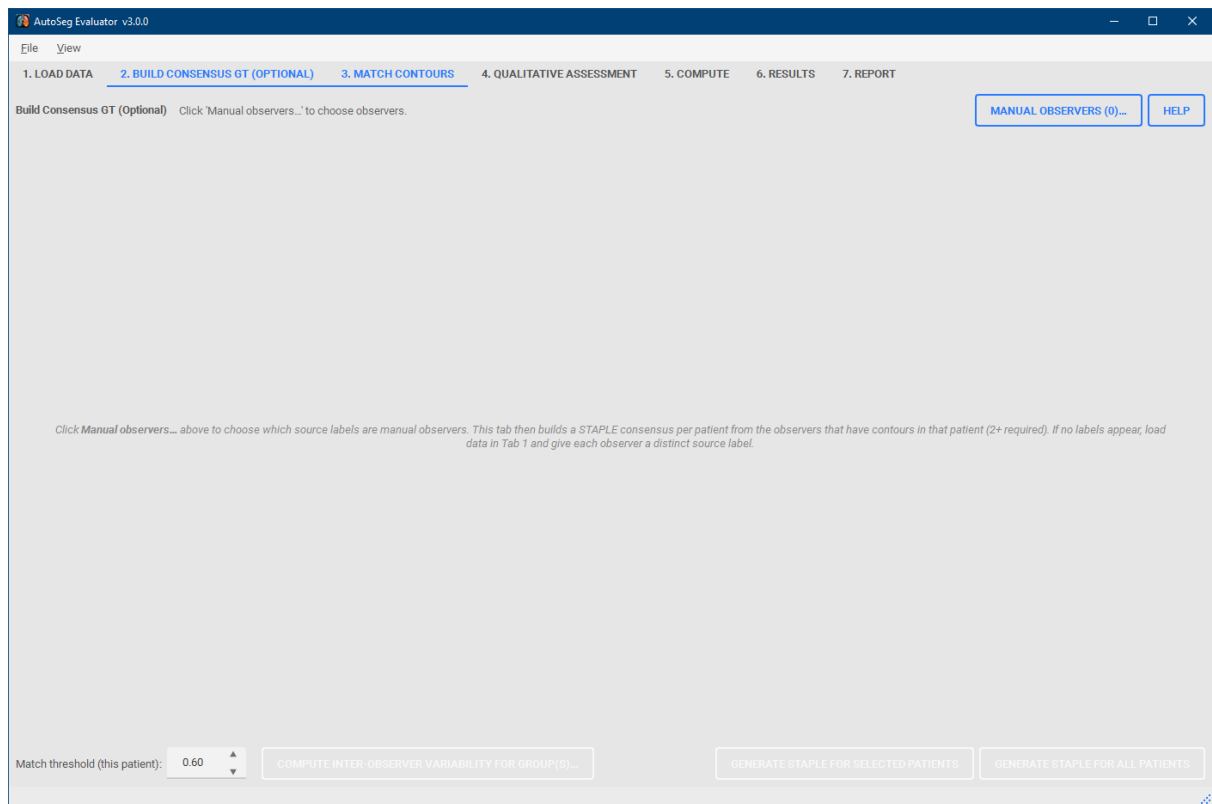
	4.1 Consensus ground truth	4.2 Drawer consensus
Question	"Build a ground truth from several manual observers, then measure contours against it."	"How well does each contour in a drawer agree with the consensus of them all?" May be useful in instances where there is no ground truth available.
Raters	The manual observers you choose	Every contour in the drawer, optionally including the ground truth
Did the scored contours help build it?	No	Yes

Set up in	Tab 2, Build Consensus GT	Tab 3, the vs STAPLE box on a drawer
Mode column in Results	Multi-observer STAPLE	Generic STAPLE with GT / no GT

4.1 Consensus Ground Truth from Several Observers

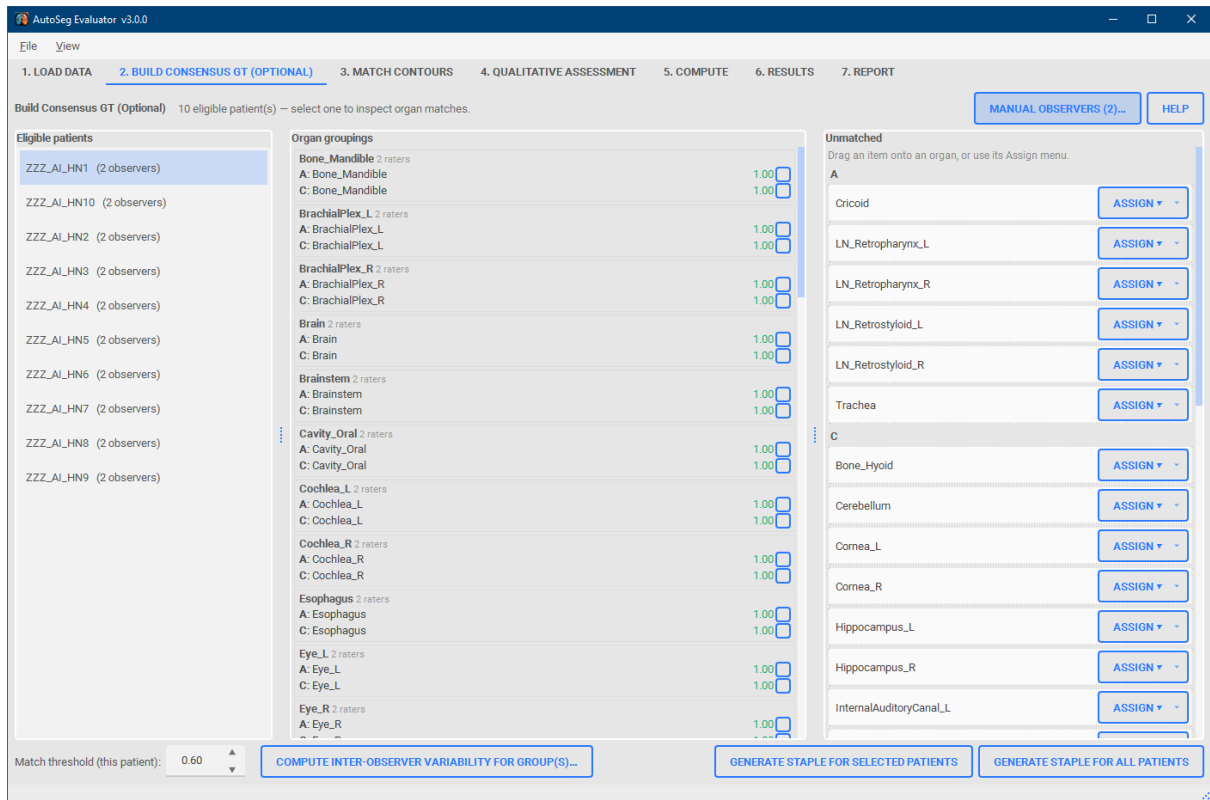
Before you start: give each observer's structure sets a distinct source label, such as Observer A and Observer B, using **Manage Source Labels...** in the Load Data tab.

- 1) In the **Build Consensus GT** tab, press **Manual Observers...** and tick the observers' labels. Every patient with two or more of these observers is listed under Eligible patients.

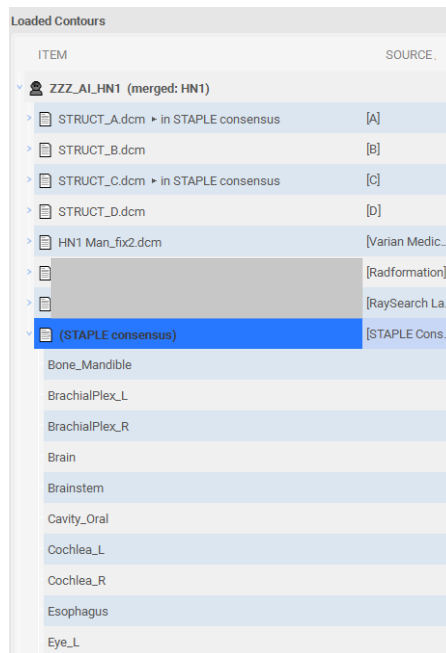


- 2) Check the organ groupings for each patient. Each observer's contours are grouped by organ name, using the same matching as Match Contours.

- Adjust **Match threshold (this patient)** if grouping is too loose or too strict. The default is 0.60.
- Contours that didn't group wait under Unmatched; use Assign ▼ to place them, or click the box to remove one from a group.
- Editing a patient by hand turns automatic grouping off for that patient. Reset turns it back on.
- An organ needs at least two observers who contoured on the same CT.



- Inter-observer variability** (optional): select patients by Ctrl + Click or Shift + Click, then click **Compute inter-observer variability for group(s)...** then select metrics. The resulting computation will produce each metric between every pair of observers, which you can copy with Copy all to clipboard (TSV) or save with **Export CSV**.
- Generate the consensus.
 - Generate STAPLE For All Patients** creates one consensus structure set per eligible patient, with source label STAPLE Consensus, and replaces any earlier ones.
- Make the STAPLE contours the ground truth. On **Match Contours**, open **Define Template...**, enter STAPLE Consensus under **Source label contains**, then click **Run Auto-Match**.



- 6) Compute as in Section 3.4. In **Results**:
 - a. Each test contour also gets STAPLE Sensitivity and Specificity;
 - b. Each organ gets a STAPLE Details row describing the consensus: consensus volume, rater disagreement, rater volume range, uncertain band, mean entropy, number of raters, iterations, and whether it converged.
- 7) In the **Report** tab: the consensus is selected as the Ground truth by default. Switch to the manual contours in the same list if you want them instead.

Limits of a consensus ground truth. It exists only as a voxel mask, so the 2D contour metrics can't be measured against it. Its dose statistics are taken from its voxels rather than from contours.

4.2 Drawer Consensus (“vs STAPLE”)

- 1) Set up Match Contours as in Section 3.
- 2) In each drawer you want, tick **vs STAPLE**.
 - a. **GT in pool** (on by default) includes the ground truth as one of the raters.
 - b. Turn it off to build the consensus from the test contours only.
- 3) After computing, every contour in the drawer is measured against the drawer's consensus and gets STAPLE sensitivity and specificity. Each organ gets a STAPLE Details row. If **vs GT** is still ticked, the usual comparisons with the ground truth are computed as well.

The contours being scored helped build this consensus, so a high score means agreement with the group, not closeness to the truth. Results are labelled STAPLE (drawer pool).

4.3 STAPLE Settings

The **STAPLE consensus parameters** on the Compute tab apply to both workflows:

Setting	Default	What it does
Max iterations	500	A ceiling, rarely reached: STAPLE usually converges well before. Check the STAPLE converged column.

Confidence weight	1.0	Biases the consensus towards (above 1) or away from (below 1) the structure. Leave at 1.0 unless you have a reason.
Adaptive bbox FG ratio max	0.50	STAPLE runs in a box around the raters' contours, not the whole scan. The box grows until the structure fills no more than this fraction of it.

5. Qualitative Workflow

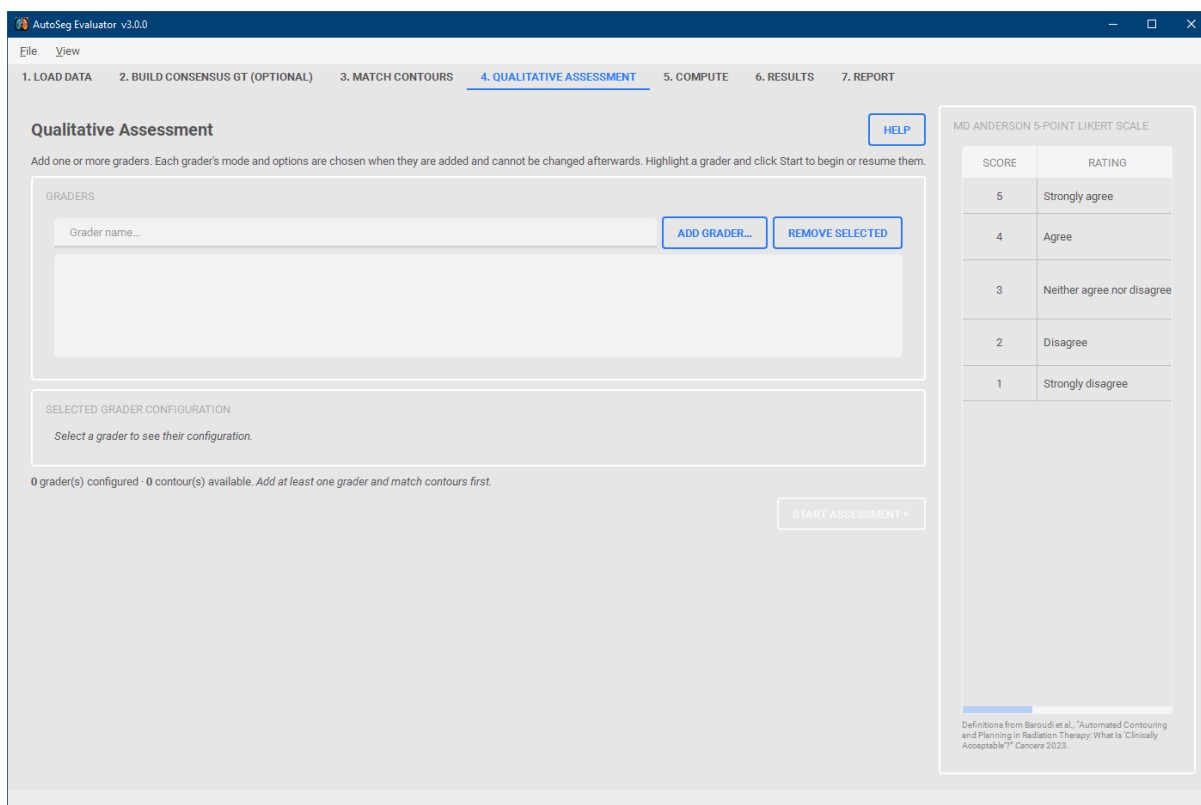
The **Qualitative Assessment** tab (Tab 4) lets one or more graders score contours on the MD Anderson 5-point Likert scale. It grades the contours matched on **Match Contours**, so no computing is needed. You can run a qualitative study on its own, or alongside the metrics in Section 3.

5.1 The Scale

Score		Meaning
5	Strongly agree	Use as is: clinically acceptable, could be used for treatment without change.
4	Agree	Minor edits that aren't necessary; stylistic differences only.
3	Neither agree nor disagree	Minor edits that are necessary, quicker than starting from scratch.
2	Disagree	Major edits; you would rather start from scratch.
1	Strongly disagree	Unusable.

5.2 Setting Up

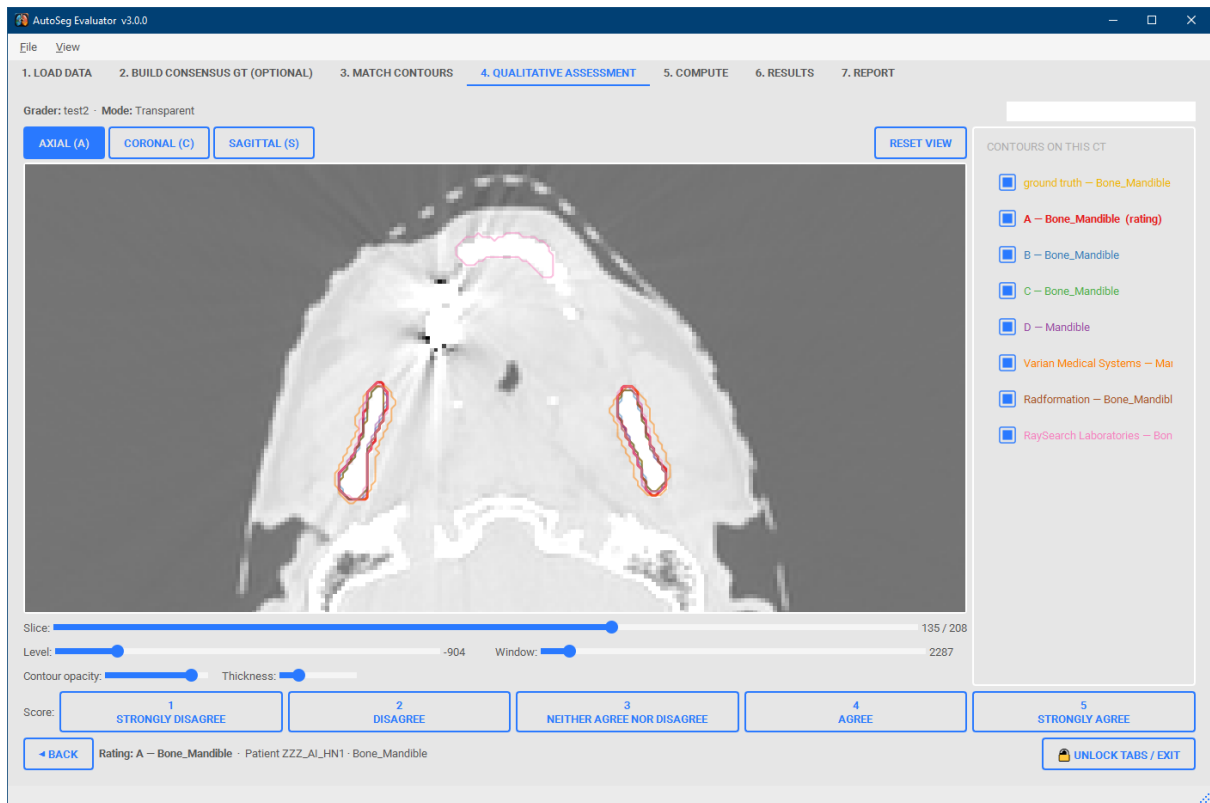
- 1) **Load Data** and **Match Contours** as in Section 3 (sections 3.2–3.3). The organ drawers decide which contours are graded.
- 2) On **Qualitative Assessment**, type a name under **Graders** and click **Add grader....**
- 3) Choose that grader's configuration. It can't be changed once the grader is added.



Option	What the grader sees
Blinded	One contour at a time, with its source hidden. Use this to score without knowing which system or observer drew the contour.
Transparent	Every source's contour for the organ on the CT, with labels and on/off toggles. The contour being graded is named below the viewer and shown in bold in the list.
Include the ground-truth contour	The ground truth is graded too.
Randomize presentation order	Contours come in random order, still grouped organ by organ.

5.3 Grading

- 1) Highlight a grader and click **Start assessment ►**. The other tabs lock until you leave, so nothing about the contours being graded can be seen elsewhere.
- 2) Inspect the contour in the viewer:
 - a. cycle Axial / Coronal / Sagittal,
 - b. Ctrl + scroll to zoom,
 - c. adjust contrast with the Level / Window sliders,
 - d. set the contours' opacity and thickness.
- 3) Click a score, 1–5. The next contour slides in.



- 4) ◀ **Back** revisits earlier contours; their scores are kept.
- 5) **Unlock tabs / exit** leaves at any time, and progress is kept.
- 6) **File** → **Save Session** stores the graders, their configurations and every score. Loading the session restores them, so grading can carry on in another sitting.
- 7) To carry on, highlight the grader and click **Start / resume assessment** ▶.
- 8) Results are stored in the **Results tab**: each grader gets a **Likert — name** column, plus a **Scored at — name** column recording when each score was given.
- 9) Scores are also displayed in the **Report tab**: scores appear in the **Metric list**, under **Qualitative (Likert)**, whether or not the contours' metrics were computed.